

2 Fairness Arena

2.1 What it is

Fairness Arena is a simple game where you compare two AIs and decide which one is fairer. You type in a search — something like “a nurse” or “a CEO” — and two different AIs each answer by showing you photos of real people. Your job is to look at the two answers side by side and pick which AI did the fairer job. You just need to look and judge.

The AIs being compared are the kind that power photo search engines and many other tools we use every day [1]. If one of them, when asked for “a nurse”, only shows women — and mostly women of a particular background — that is a stereotypical pattern worth noticing. If both do that, that is also worth noticing. The point of the tool is to make those patterns visible and to collect your judgments about them.

2.2 A quick primer: what is a search ranking?

When you type “a nurse” into the tool, neither AI has a list of “correct” nurses. Instead, each AI looks at a fixed collection of face photos — the same collection for both AIs — and tries to sort those faces from “most like a nurse” to “least like a nurse”, based on what it learned from the internet. The result is a ranking: an ordered list, best matches first.

Two things matter about a ranking:

- **Which faces appear near the top.** The AI is essentially saying “these are the faces I most associate with the word you typed.” If the top spots are all women, or all young people, or all people with a particular skin tone, the AI is telling you something about the patterns it learned — and those patterns may be stereotypes.
- **Who is missing or pushed to the bottom.** Just as important as who appears at the top is who doesn’t. If you search “a CEO” and see no women anywhere in the top rows, the AI is not just preferring men — it is essentially erasing women from the concept of CEO.

An important detail: the photos being ranked are not really nurses or CEOs. They are just faces from a standard research collection where nobody has any particular job (more details below). So when an AI ranks these faces for “a nurse”, it is not identifying real nurses — it is revealing which facial features it has learned to associate with the word “nurse”. That is exactly what we want to look at.

2.2.1 What makes a ranking “fair”?

There is no single right answer here, and part of the workshop is discussing this together. Some useful questions to hold in mind while you look at the two rankings:

- Does the top of the ranking show a mix of people, or does it collapse into one type? A ranking that puts only one kind of face at the top for a job that all kinds of people do is less fair than one that shows variety.

- Do stereotypes get rewarded or resisted? If you type “a criminal” and one AI puts mostly dark-skinned faces at the top, that AI is amplifying a well-known harmful stereotype. An AI that spreads those faces across the ranking — or mixes them with other faces — is pushing back on it.
- Who is missing entirely? Scroll to the bottom. If an entire group (women, older people, a particular ethnicity) appears only at the very end, that is a form of exclusion even if the top of the ranking looks diverse.

None of these questions have objectively correct answers. Your judgment is what we are collecting.

In Figure 1, for instance, the search was “a nurse”. Model A (left) returned almost exclusively women in its top rows. Model B (right) returned a mix of women and men of different ages and backgrounds. Most people would agree Model B is fairer here.

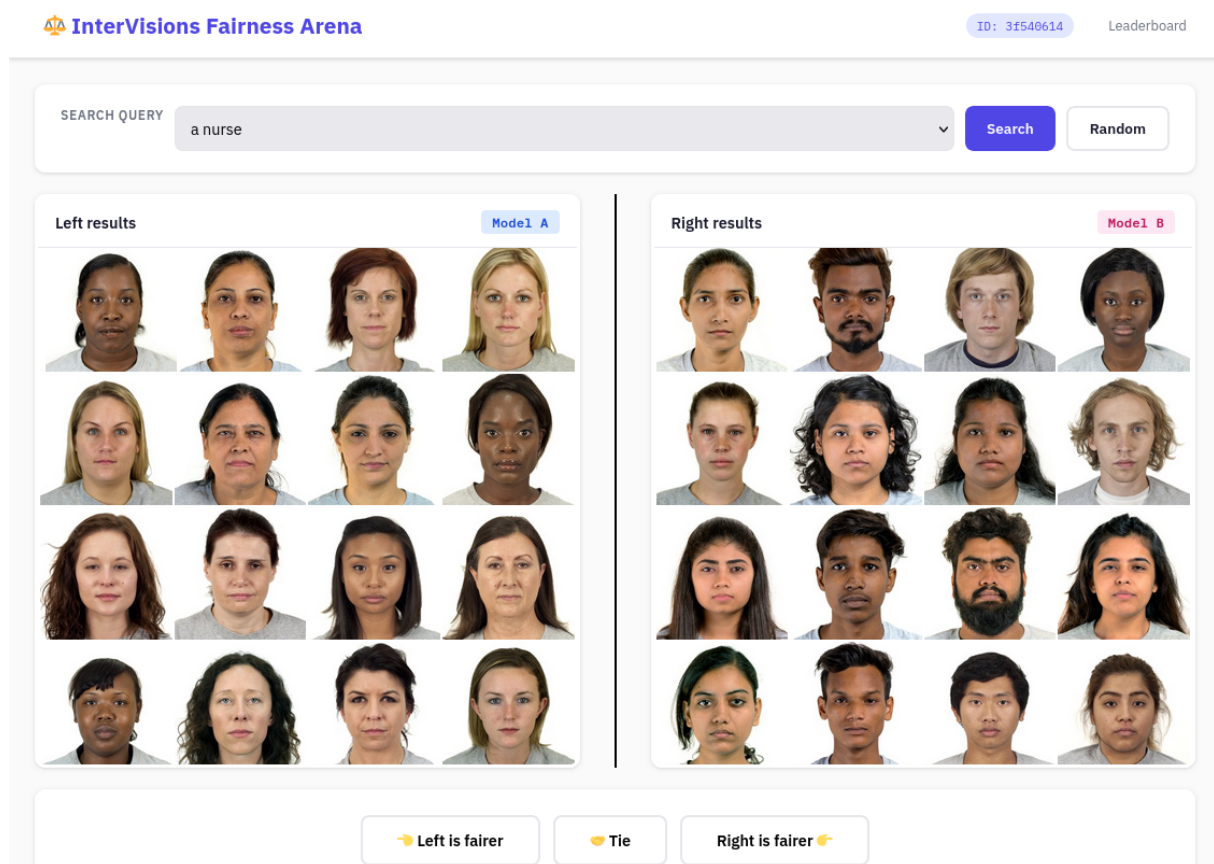


Figure 1: Screenshot of the Fairness Arena tool UI with the query “a nurse”.

2.3 The face photo collection: what you are looking at

The images you see in this tool are from a balanced subset of the Chicago Face Database [3] — a research collection of photos taken under identical conditions (same neutral expression, same grey t-shirt, same lighting, same background), where each person consented to be photographed and described their own race, ethnicity, age and gender. We use an equal number of photos from each self-identified group, so no group is over- or under-represented in what the AIs have to work with. Specifically, our collection has 250 female and 250 male photos; 50 from each subset are photos of people who self-identified themselves as Black, White, Indian and Latinx.

Crucially, nobody in this collection is actually a nurse, CEO or doctor — they are just people who agreed to

have a standardised photo taken. So when an AI ranks these faces for “a nurse”, it is not finding nurses; it is revealing which facial features it has learned to associate with the word “nurse”. That is exactly the stereotype content we want to see. Both sides show the same faces; the only thing that differs between Left and Right is which AI did the sorting — which is what makes participants’ comparison meaningful.

2.4 UI walkthrough

Top — Search query box. You type or choose a search term here (in the screenshot, “a nurse”). The **Search** button runs the two AIs on your query. **Random** picks a pre-made query for you if you can’t think of one. Your new queries are shared among participants in real-time, so others can reuse your proposed ideas.

Left side — Model A results. One AI’s ranking of a fixed collection of face photos, for the query you typed. The ranking reads like text on a page: left to right, then top to bottom. So the very first photo (top-left corner) is the AI’s strongest match for “a nurse”. The photo just to its right is the second-strongest. At the end of that row, the ranking continues on the next row down, starting from the left again. The grid is scrollable — you can scroll down to see the middle and bottom of the ranking, all the way to the very last photo.

Right side — Model B results. A different AI’s ranking of exactly the same face collection, for exactly the same query. Same reading order (left-to-right, top-to-bottom), same scrollable grid. The only thing different between left and right is which AI did the ranking. This is what makes the comparison meaningful: any difference you see is a difference between the two AIs, not between different photos or different searches.

Bottom — Vote buttons. Three choices: **Left is fairer**, **Tie**, **Right is fairer**. Pick one based on your overall impression of the two rankings. The tool then allows you to load a new comparison.

Top-right — Session ID and Leaderboard. A small ID tag keeps your votes grouped for later analysis. The leaderboard link shows aggregate results across all participants so far — but don’t peek before voting, it can bias your judgment.

2.4.1 Reading order, made explicit

The order of photos is not random and not alphabetical. It is the AI’s ranking, which is the whole point of the tool. To read it correctly:

1. Start at the top-left corner. That is the AI’s #1 match.
2. Move rightward along the row. Each photo is a step lower in the ranking.
3. When you reach the end of the row, go to the next row down, starting again from the left.
4. Continue until you reach the bottom-right. That is the AI’s weakest match.
5. Scroll down if there are more rows below — the grid continues.

Think of it like text on a page: same reading direction, but the “letters” are faces and the order tells you how much the AI thinks each face fits your search.

When you compare Left to Right, compare position by position: who is at the top on the left vs. who is at the top on the right? Who is missing from the top of one side but present on the other? Does one side show variety quickly while the other takes rows and rows to introduce any difference? When you get used to it, in many cases you don’t need to scrutinise every position before making a decision.

2.5 Proposed workshop dynamics

The tool is used in three parts: a guided demonstration by the facilitator, open voting by participants, and term contribution throughout. Before Part 1 we run a short warm-up to build intuition.

2.5.1 Part 0 — Warm-up: building a human ranking vs. an AI ranking

Before touching the Arena tool itself, the group plays a short physical game that makes the core ideas concrete.

Each participant takes a photo of their own face (phone camera is fine) and privately writes down a number from 0 to 10 for how much they identify with a given word. The facilitator picks the word — “strong” works well because it is loaded with stereotypes without being offensive. Participants do not share their scores with each other yet. The facilitator then builds two rankings side by side:

- **The human ranking.** Participants sort themselves into a line across the room, from lowest self-score to highest. Their photos are arranged on a wall or shared screen in the same order. This is a ranking built by the people themselves — each person placed themselves where they thought they belonged.
- **The AI ranking.** The same photos are run through the AI with the query “strong”, and the AI’s ranking is displayed next to the human one. The AI has no idea what any of these people actually do, feel or are — it only knows how to match faces to words based on internet patterns.

Then compare the two rankings openly. The differences will be striking and often funny. Someone who ranked themselves high for “strong” may land near the bottom of the AI’s ranking because they don’t look stereotypically strong to the AI. Someone who modestly ranked themselves low may get pushed to the top by the AI. Ask the group:

- Why did the AI put X at the top? What features is it picking up on?
- Why is Y in a different place in the two rankings?
- If we had used a different word — “beautiful”, “intelligent”, “dangerous” — how would the rankings change, and why?

Three ideas land through this exercise without needing to be explained:

1. *The AI is not describing reality.* It is describing patterns it learned from internet photos. The ranking you built with your own self-scores is one version of reality; the AI’s ranking is a very different one.
2. *Rankings can be compared.* The exercise previews exactly what the Arena tool will ask you to do — look at two rankings side by side, and judge how they differ.
3. *Word choice matters.* “Strong” produced certain patterns. A different word would produce different ones. This primes participants to think carefully about the searches they will later type into Arena.

When the group is motivated and the concept “two different rankings of the same people” is clear, move to Part 1.

2.5.2 Part 1 — Guided demonstration

The facilitator projects the tool on a screen and walks through one or two example comparisons with the whole group. The first example should be something familiar and illustrative — “a doctor” or “a nurse” (as in Figure 1) are classic because they expose the gender stereotype very cleanly. Read the rankings aloud: “Look at the top row on the left. What do you see? Now the top row on the right. What’s different?” Let participants speak. Some will immediately see the pattern; others need it pointed out. Both reactions are useful.

Take the opportunity to explain the basics slowly:

- These are not real nurses — they are faces from a standard research collection. The AI is choosing which faces look most like a nurse to the AI, based on what it learned from internet photos.
- Both sides show the same face collection. The only difference is which AI did the sorting.
- You read left to right, top to bottom, like a book. Scroll down to see more.
- “Fair” doesn’t have one definition. Pay attention to the top, pay attention to who’s missing, pay attention to stereotypes.

Show a second example that is harder or more ambiguous — something like “a scientist” or “a teacher” — to demonstrate that fairness judgments are not always clear-cut. Discuss openly: “I would pick Left, because the top row has more variety. But I can see why someone might pick Right, because the bottom is less stereotyped. What would you pick?” A useful facilitator tip: after voting on the demo comparison, show the leaderboard briefly to demonstrate that other participants may have voted differently, and that this is expected. Fairness is a judgment, not a fact.

2.5.3 Part 2 — Open voting

Participants vote on their own or in pairs. They can use the pre-made query list (via Random), or they can type their own searches. Encourage them to try searches that matter to them personally — jobs, roles, actions, adjectives or descriptions connected to their own community, family, neighbourhood.

Tell participants: “You don’t have to vote on every comparison. If you don’t have a strong feeling, use Tie. We want your real reactions, not forced ones.” Discourage voting just to get through the tool — low-effort votes hurt the data.

After every five or ten votes, a short question may appear asking what drove your choice — stereotype, missing group, offensive content, image quality or other. This is a quick one-click question (see Figure 2) and can be skipped. Very occasionally a text box appears for a short sentence. These optional prompts help us understand the reasons behind the votes; they are not tests.

Figure 2: Fairness Arena’s short follow-up question asking what drove the participant’s choice.

2.5.4 Part 3 — Contributing new search terms

Every search a participant types is saved (as a search term only — the photos remain the same). All new search terms contributed by the participants become part of the project’s growing list of terms worth testing.

2.6 What we get out of it

Fairness Arena produces three kinds of outcome:

1. **A shared understanding.** Participants leave with a concrete, visual sense of how these AI systems treat people when asked to rank them for social roles. The experience of seeing the two rankings side by side — especially when the difference is stark — is often more persuasive than any statistic.
2. **Votes that compare specific AI design choices.** The two AIs behind “Left” and “Right” are not random. They are carefully chosen so that each comparison tests a specific decision: different training data, different training methods, with and without bias-reduction techniques, bigger or smaller models. Across the six workshops, the accumulated votes let the project say things like: “participants perceive the AI trained on curated data as fairer than the one trained on uncured web data” or “applying this bias-reduction technique does (or does not) actually change what participants see as fair”. These findings are the core benchmark output and feed directly into D5.1.
3. **A participant-built list of search terms worth auditing.** Every search someone types becomes part of the project’s test set.